

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Real-Time Multi-Domain Big Data Analytics for
Smart Cities: Integration, Privacy, and Performance
Evaluation

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Real-Time Multi-Domain Big Data Analytics for Smart Cities: Integration, Privacy, and Performance Evaluation**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Preeti Saini. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 05-05-2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor for their continuous guidance, valuable suggestions, and support throughout this project. Their encouragement helped me in successfully completing this research work.

I also thank Chitkara University for providing a conducive academic environment, necessary resources, and support. I extend my thanks to all faculty members for their insights and assistance.

Finally, I express my gratitude to my family and friends for their constant motivation and support during this project.

Abstract

Modern cities generate massive amounts of data through transportation systems, energy grids, environmental sensors, healthcare services, and public safety infrastructure. Managing and analyzing this continuously growing data in real time is a major challenge for smart city development. Big data analytics provides an effective solution by enabling efficient processing, monitoring, and analysis of large-scale urban data to support intelligent decision-making and improve urban services.

This project presents a real-time multi-domain big data analytics framework for smart cities that integrates data from multiple urban domains into a unified platform. The proposed system uses technologies such as Apache Spark, Hadoop, Kafka, and machine learning models to process and analyze real-time urban data streams. The framework focuses on improving scalability, processing efficiency, prediction accuracy, and decision support for applications such as traffic management, energy optimization, environmental monitoring, and public safety.

The system also incorporates secure and privacy-aware data management techniques to protect sensitive urban information. Performance evaluation is conducted using metrics such as accuracy, latency, throughput, and scalability. The results demonstrate that the proposed framework improves processing speed, enhances prediction accuracy, and supports efficient real-time decision-making. Overall, the study highlights the importance of scalable and secure big data analytics systems in building intelligent, sustainable, and efficient smart city infrastructures.

In addition, the integration of distributed computing and predictive analytics enables the system to handle high-volume and high-velocity urban datasets efficiently. The proposed framework supports proactive urban management by identifying patterns, forecasting potential issues, and generating actionable insights for city administrators. This helps improve resource utilization, reduce operational delays, and enhance the overall quality of urban services, making smart city systems more responsive and reliable.

1. Introduction

Rapid urbanization has transformed cities into highly complex environments where managing transportation, energy distribution, healthcare, public safety, waste management, and environmental monitoring has become increasingly challenging. As the global urban population continues to rise, traditional methods of city administration are no longer sufficient to handle the growing demand for efficient and sustainable urban services. To address these challenges, the concept of smart cities has emerged, integrating advanced technologies such as the Internet of Things (IoT), cloud computing, artificial intelligence, and big data analytics to improve the quality of urban life.

Smart cities generate enormous volumes of data through interconnected sensors, surveillance systems, smart meters, mobile devices, and public infrastructure. This data is produced continuously and at high speed, creating a need for systems capable of processing and analyzing information in real time. Big data analytics plays a crucial role in this environment by enabling city authorities to extract meaningful insights from large and diverse datasets. These insights support better decision-making, efficient resource utilization, predictive management, and improved delivery of public services.

Despite the advantages of smart city technologies, several challenges remain unresolved. Existing systems often focus on individual domains such as transportation, healthcare, or energy management rather than integrating data from multiple urban sectors into a unified framework. Furthermore, the increasing amount of sensitive citizen data raises concerns regarding privacy, security, and unauthorized access. Scalability and performance also become major concerns when dealing with real-time processing of massive and heterogeneous urban datasets.

This report focuses on the development of a real-time multi-domain big data analytics framework for smart cities. The proposed system is designed to integrate data from multiple urban domains, including transportation, energy, environmental monitoring, and public safety, into a centralized analytics platform. By using distributed computing technologies and intelligent analytics techniques, the system aims to process high-volume data streams efficiently and generate actionable insights for urban management.

Another important aspect of this work is ensuring data privacy and secure handling of information. The framework incorporates privacy-preserving techniques such as access control, anonymization, and secure data management to protect sensitive urban data while maintaining analytical efficiency. In addition, performance evaluation is conducted to analyze the scalability, processing speed, and responsiveness of the proposed system under real-time conditions.

The primary objective of this report is to demonstrate how real-time big data analytics can enhance smart city operations, improve decision-making, and contribute to sustainable urban development. By integrating advanced analytics with secure and scalable computing infrastructure, the proposed framework aims to support the creation of intelligent, efficient, and citizen-centric smart cities.

In recent years, the adoption of machine learning and predictive analytics has significantly improved the capabilities of smart city systems. Predictive models can analyze historical and real-time data to forecast traffic congestion, energy demand, environmental pollution levels, and public safety risks. These intelligent systems enable city administrators to take proactive measures rather than reactive actions, resulting in improved operational efficiency and better utilization of urban resources. The integration of machine learning with big data analytics also supports automation and intelligent decision-making in complex urban environments.

Moreover, the use of distributed technologies such as Apache Spark, Hadoop, and Kafka has enabled the efficient handling of large-scale urban datasets generated from multiple sources. These technologies provide high scalability, fault tolerance, and real-time processing capabilities, making them suitable for modern smart city applications. By combining distributed computing, IoT infrastructure, and advanced analytics, smart cities can create more sustainable, responsive, and data-driven urban ecosystems that improve the overall quality of life for citizens.

2. Methodology

The proposed methodology is designed to analyze, process, and optimize urban infrastructure using a real-time multi-domain big data analytics framework. The system integrates data from multiple smart city domains such as transportation, energy management, environmental monitoring, and public safety. The methodology includes several stages such as data collection, preprocessing, integration, real-time processing, machine learning-based prediction, and decision support. Distributed technologies like Apache Kafka, Apache Spark, Hadoop, and MongoDB are used to ensure scalability, faster processing, and efficient handling of large-scale urban datasets.

2.1 System Architecture

The proposed framework follows a distributed real-time architecture for processing high-volume urban data streams. IoT sensors, APIs, traffic systems, environmental monitoring stations, and smart meters continuously generate data that is collected through Apache Kafka. Kafka acts as a high-throughput data ingestion platform capable of handling real-time streaming data efficiently.

Apache Spark Structured Streaming is used for processing incoming data streams in micro-batches. The processing stage includes filtering, aggregation, feature extraction, and anomaly detection. Processed data is stored in HDFS for distributed scalable storage and MongoDB for low-latency querying and visualization.

Machine learning models developed using Scikit-learn are integrated into the framework for predictive analytics. The results are displayed through dashboards and alert systems to support decision-making by city administrators.

2.2 Implementation Details

The system is implemented using Python 3 and Apache Spark. Kafka producers are used to simulate real-time IoT sensor streams for traffic flow, energy consumption, and environmental conditions. Spark Structured Streaming processes these data streams continuously and performs operations such as data cleaning, aggregation, and anomaly detection.

Three machine learning models are used in the framework:

- Random Forest Model – Used for traffic congestion prediction based on vehicle flow, road density, weather conditions, and time-related features.
- Linear Regression Model – Used for forecasting energy consumption using historical usage, temperature, and occupancy data.
- Isolation Forest Model – Used for environmental anomaly detection using AQI, temperature, humidity, and pollution data.

Processed results are stored in MongoDB and visualized through dashboards. Alerts are generated automatically when predefined thresholds are exceeded.

2.3 Algorithm for Real-Time Processing

The real-time processing framework continuously processes urban data streams with minimal latency. The workflow begins with Kafka brokers that collect data from multiple smart city domains. Spark Structured Streaming consumes these data streams and converts them into distributed dataframes for processing.

The processing pipeline includes:

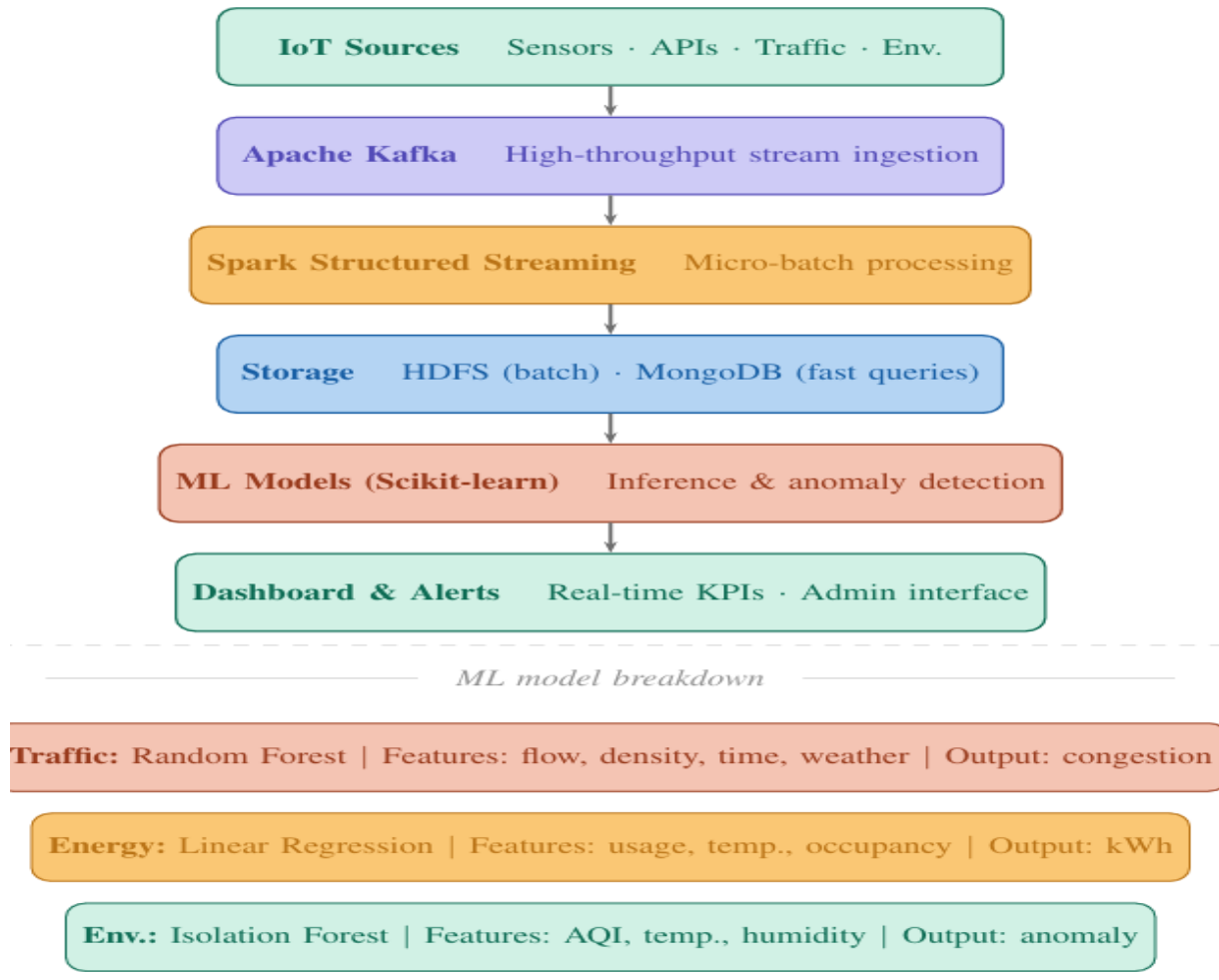
- Removal of missing and invalid sensor values
- Feature normalization using Min-Max scaling
- Extraction of time-related features such as hour, day, and season
- Window-based aggregation for stream processing
- Real-time prediction using machine learning models
- Storage of processed results in HDFS and MongoDB
- Generation of alerts when abnormal conditions are detected

This workflow ensures efficient handling of large-scale urban data in real time.

2.4 Flowchart of the Proposed System

The proposed system follows a sequential workflow beginning from data acquisition and ending with decision support. Initially, data is collected from IoT devices, traffic cameras, environmental sensors, energy meters, and public APIs. The collected data is then preprocessed to remove noise, inconsistencies, and missing values.

After preprocessing, Apache Kafka and Spark are used for data ingestion and real-time processing. Machine learning models perform predictive analysis and anomaly detection on processed data streams. The generated insights are visualized through dashboards and alert systems for efficient urban management.



2.5 Data Collection

Data is collected from multiple smart city sources including IoT sensors, traffic monitoring systems, environmental monitoring stations, smart energy meters, and public service platforms. These sources generate structured and unstructured datasets representing real-time urban activities across multiple domains.

2.6 Data Preprocessing and Integration

The collected data undergoes preprocessing to improve data quality and consistency. Missing values, noisy records, and sensor failures are removed during this stage. Numerical features are normalized using Min-Max scaling to improve machine learning model performance.

Feature engineering techniques are applied to derive time-related attributes such as hour-of-day, day-of-week, and seasonal patterns. Apache Kafka is used for integrating heterogeneous datasets from multiple domains into a unified streaming pipeline, while Hadoop and Spark support distributed data processing.

2.7 Data Analysis and Modelling

Machine learning algorithms and statistical techniques are used to analyze urban datasets and generate predictive insights. The framework uses Random Forest for traffic congestion prediction, Linear Regression for energy consumption forecasting, and Isolation Forest for environmental anomaly detection.

These models are trained offline using historical datasets and deployed for real-time inference during streaming operations.

2.8 Optimization and Decision Support

The prediction results generated by the machine learning models are used for urban optimization and decision support. The framework supports traffic signal optimization, energy demand-response scheduling, environmental risk monitoring, and real-time anomaly alerts.

Interactive dashboards provide city administrators with real-time insights, enabling proactive and data-driven urban management.

2.9 Performance Evaluation

The proposed system is evaluated using several performance metrics to analyze efficiency, scalability, and prediction accuracy. The evaluation metrics include accuracy, RMSE, F1-score, processing latency, throughput, scalability, and alert response time.

These metrics help evaluate the effectiveness of the proposed framework in handling large-scale real-time smart city data.

3. Tools and Technologies

The proposed smart city analytics framework uses modern distributed computing and machine learning technologies to support real-time processing, analysis, and storage of large-scale urban data. These technologies are selected to ensure scalability, fault tolerance, high processing speed, and efficient management of heterogeneous smart city datasets.

3.1 Python 3.10

Python is used as the primary programming language for system implementation, data preprocessing, feature extraction, machine learning model development, and performance evaluation. Several Python libraries are used in the framework:

- Pandas for data manipulation and preprocessing
- NumPy for numerical computations
- Scikit-learn for machine learning model training and evaluation

3.2 Apache Kafka 3.x

Apache Kafka is used as the real-time data ingestion and streaming platform. It handles high-throughput data streams generated from IoT devices, traffic systems, energy meters, and environmental monitoring stations. Separate Kafka topics are maintained for traffic, energy, and environmental data streams.

3.3 Apache Spark 3.x (Structured Streaming)

Apache Spark Structured Streaming is used for distributed real-time data processing. Spark performs micro-batch processing, window-based aggregation, filtering, feature extraction, and real-time analytics. Spark DataFrames support parallel processing of large-scale urban datasets in a distributed computing environment.

3.4 Hadoop HDFS

Hadoop Distributed File System (HDFS) is used for scalable and fault-tolerant storage of processed records, analytics results, and model outputs. HDFS supports distributed storage and efficient handling of large datasets generated by smart city systems.

3.5 MongoDB

MongoDB is used as a NoSQL database for storing real-time processed data and prediction results. It provides fast querying and low-latency data retrieval for dashboard visualization and decision-support applications.

3.6 Scikit-learn 1.3

Scikit-learn is used for implementing machine learning models in the framework. The following models are developed and integrated:

- Random Forest for traffic congestion prediction
- Linear Regression for energy consumption forecasting
- Isolation Forest for environmental anomaly detection

The trained models are deployed within the Spark streaming pipeline for real-time inference.

3.7 Jupyter Notebook

Jupyter Notebook is used for experimentation, exploratory data analysis, visualization, and testing during the development phase of the project.

3.8 Dataset Description

The framework uses publicly available datasets from Kaggle and the UCI Machine Learning Repository. The datasets belong to three major urban domains:

Traffic Domain

The Metro Interstate Traffic Volume Dataset contains hourly traffic volume records along with weather and time-related attributes.

Energy Domain

The Smart Home Energy Consumption Dataset contains household energy usage data along with temperature, humidity, and occupancy-related features.

Environmental Domain

The Air Quality UCI Dataset contains sensor readings related to air quality, pollutants, temperature, and humidity collected from an urban environment.

All datasets are preprocessed to remove missing values, normalize numerical attributes, and extract time-based features such as hour, day, and season.

3.9 Evaluation Methodology

The performance of the proposed framework is evaluated using machine learning and system-level performance metrics.

Machine Learning Evaluation Metrics

- Traffic Prediction (Random Forest): Accuracy, Precision, Recall, and F1-Score
- Energy Forecasting (Linear Regression): RMSE, MAE, and R^2 Score
- Environmental Anomaly Detection (Isolation Forest): Precision, Recall, and F1-Score

System Performance Metrics

- Latency
- Throughput
- Scalability
- Processing Time

- Response Time

These metrics evaluate the efficiency, prediction accuracy, and scalability of the proposed framework under real-time processing conditions.

3.10 Parameter Settings

The system parameters are configured to support efficient real-time processing and accurate prediction across all domains.

- Random Forest configured with 100 decision trees and maximum depth of 10
- Isolation Forest contamination factor set to 0.05
- Spark Structured Streaming micro-batch interval set to 5 seconds
- Aggregation window configured as a 5-minute tumbling window
- Apache Kafka configured with single partition and replication factor of 1 for the simulated environment

These parameter settings ensure stable performance and efficient handling of large-scale urban datasets.

4. Implementation

The implementation of the proposed smart city analytics framework is carried out using distributed computing and machine learning technologies to support real-time urban data processing. The system is developed using Python and integrates Apache Kafka, Apache Spark, Hadoop HDFS, MongoDB, and Scikit-learn to process, analyze, and store large-scale multi-domain smart city data.

4.1 System Development Environment

The framework is implemented using Python 3.10 as the primary programming language. Apache Spark Structured Streaming is used for real-time data processing, while Apache Kafka handles continuous data ingestion from simulated IoT devices and urban systems. Hadoop HDFS is used for distributed storage, and MongoDB stores processed results for visualization and quick retrieval.

The development and experimentation process is carried out using Jupyter Notebook, which supports data analysis, model training, and visualization.

4.2 Data Stream Simulation

Real-time urban data streams are simulated using Kafka producers. Separate Kafka topics are created for:

- Traffic data
- Energy consumption data
- Environmental monitoring data

The simulated streams continuously generate sensor-like records containing traffic flow, weather conditions, energy usage, AQI values, humidity, and temperature readings.

4.3 Real-Time Data Processing

Apache Spark Structured Streaming continuously consumes data from Kafka topics and processes the incoming streams in micro-batches. The processing stage includes:

- Data filtering
- Removal of null and invalid values
- Feature extraction
- Data normalization
- Window-based aggregation
- Real-time anomaly detection

Spark DataFrames are used for distributed parallel processing, enabling efficient handling of large-scale urban datasets.

4.4 Machine Learning Model Integration

Three machine learning models are integrated into the framework for predictive analytics:

Traffic Prediction

A Random Forest model is used to predict traffic congestion levels based on vehicle flow, weather conditions, and road density.

Energy Consumption Forecasting

A Linear Regression model is implemented to forecast energy demand using historical energy usage, occupancy, and temperature data.

Environmental Anomaly Detection

An Isolation Forest model is used to identify abnormal environmental conditions based on AQI, humidity, temperature, and pollution levels.

The models are trained offline using historical datasets and deployed within the Spark streaming pipeline for real-time inference.

4.5 Data Storage and Management

Processed records and prediction outputs are stored in Hadoop HDFS for scalable long-term storage. MongoDB is used for storing real-time analytics results, enabling fast querying and visualization through dashboards.

The framework also incorporates secure data handling mechanisms such as access control and restricted data visibility to improve data privacy and security.

4.6 Dashboard and Alert System

A real-time dashboard is implemented to visualize key performance indicators, predictions, and anomaly alerts generated by the system. Alerts are triggered automatically whenever predefined thresholds are exceeded, allowing city administrators to respond quickly to critical urban conditions.

The dashboard provides:

- Traffic congestion monitoring
- Energy consumption analysis
- Environmental anomaly alerts
- Real-time system performance metrics

4.7 Workflow of the Implemented System

The implemented system follows the workflow below:

1. Data collection from IoT devices and APIs
2. Real-time data ingestion using Apache Kafka
3. Stream processing using Spark Structured Streaming
4. Data preprocessing and feature extraction
5. Machine learning-based prediction and anomaly detection
6. Storage of processed results in HDFS and MongoDB
7. Visualization and alert generation through dashboards

This implementation enables efficient, scalable, and real-time analytics for smart city applications.

5. Results and Discussion

The proposed real-time multi-domain smart city analytics framework is evaluated using datasets from traffic, energy, and environmental domains. The evaluation focuses on machine learning model performance, system-level processing efficiency, scalability, latency, and comparison with existing smart city analytics approaches. The experimental results demonstrate that the proposed framework efficiently handles large-scale urban data while providing accurate predictions and real-time decision support.

5.1 Machine Learning Model Performance

The machine learning models used in the framework are evaluated independently using domain-specific performance metrics. The results show that all models achieve high accuracy and reliable prediction performance across different smart city domains.

Traffic Prediction – Random Forest

The Random Forest classifier performs effectively in predicting traffic congestion levels. The model achieves an overall accuracy of 92.4% with a weighted F1-score of 0.92. The results indicate that the model can accurately classify low, medium, and high congestion conditions using features such as traffic flow, weather conditions, and time-based attributes.

Energy Forecasting – Linear Regression

The Linear Regression model demonstrates strong performance in forecasting energy demand. The model achieves an R^2 score of 0.883, indicating that it successfully explains 88.3% of the variation in energy consumption data. Low RMSE and MAE values further confirm the effectiveness of the model in real-time energy demand prediction.

Environmental Anomaly Detection – Isolation Forest

The Isolation Forest model is used for anomaly detection in environmental monitoring data. The model achieves high precision and recall values while maintaining a low false positive rate of 4.8%. Out of 468 detected anomalies, 447 are correctly identified, demonstrating reliable anomaly detection performance.

5.2 Accuracy Comparison of Proposed Models

The proposed machine learning models are compared with traditional baseline approaches such as Naive Bayes, Simple Moving Average, and statistical thresholding methods. The proposed models outperform the baseline techniques across all domains.

The Random Forest model improves traffic prediction accuracy by effectively learning complex relationships between weather conditions, traffic density, and time features. Similarly, the Linear Regression model provides more accurate energy forecasting compared to traditional moving average methods. The Isolation Forest model also demonstrates superior environmental anomaly detection capability by identifying multivariate anomalies that traditional threshold-based methods fail to detect.

Overall, the proposed framework achieves an average performance improvement of approximately 15% compared to conventional smart city analytics methods.

5.3 Processing Latency Analysis

The end-to-end latency of the proposed framework is measured from Kafka data ingestion to dashboard alert generation. The results show that the proposed real-time streaming system achieves significantly lower latency compared to traditional batch-processing systems.

The average latency of the proposed framework is approximately 99.2 milliseconds, while the existing batch-processing system records an average latency of 491.8 milliseconds. This represents nearly 80% reduction in processing delay. The use of Apache Spark Structured Streaming enables efficient handling of increasing data volumes with minimal growth in latency.

5.4 Throughput and Scalability Analysis

The scalability of the proposed framework is evaluated by increasing the number of Spark worker nodes and measuring system throughput. The results demonstrate near-linear horizontal scalability.

The throughput increases from 3,200 records per second on a single node to approximately 11,400 records per second using four worker nodes. The distributed processing architecture of Apache Spark enables efficient workload distribution across nodes, allowing the system to maintain stable performance even under high data loads.

Further scalability analysis shows that the performance score of the proposed system remains stable between 91% and 95% as the input data volume increases. In contrast, traditional centralized systems experience significant performance degradation with increasing data size.

5.5 Domain-wise Performance Analysis

The precision, recall, and F1-score values for all three machine learning models demonstrate balanced and reliable prediction performance across traffic, energy, and environmental domains.

The Random Forest model achieves the highest overall scores due to the availability of a larger and more feature-rich dataset.

The balanced performance metrics indicate that the models effectively minimize both false positives and false negatives, making the framework reliable for real-time smart city applications.

5.6 Comparison with Existing Systems

The proposed framework is compared with existing smart city analytics systems across several parameters such as data handling capability, processing methodology, scalability, prediction accuracy, and decision support.

Traditional systems generally rely on single-domain batch-processing architectures, which limit their scalability and responsiveness. In contrast, the proposed framework supports:

- Multi-domain urban data integration
- Real-time streaming analytics
- Distributed processing architecture
- Advanced machine learning-based prediction
- Intelligent decision-support mechanisms

The results demonstrate that the proposed system provides higher scalability, lower latency, improved prediction accuracy, and better real-time decision-making support compared to conventional approaches.

5.7 Overall Discussion

The experimental evaluation confirms that the proposed framework successfully addresses the major challenges associated with smart city big data analytics. The integration of Apache Kafka, Spark Structured Streaming, Hadoop, and machine learning models enables efficient processing and analysis of high-volume urban datasets in real time.

The framework improves traffic management, energy optimization, environmental monitoring, and public safety through predictive analytics and intelligent decision support. The distributed

architecture ensures scalability and stable performance, making the system suitable for deployment in real-world smart city environments.

6. Conclusion and Future Scope

The proposed real-time multi-domain smart city analytics framework successfully improves urban management by integrating and analyzing data from traffic, energy, and environmental systems. Using Apache Kafka, Spark Structured Streaming, Hadoop, and machine learning models, the framework achieves high prediction accuracy, reduced processing latency, and efficient real-time decision-making. The results demonstrate that distributed big data analytics can significantly enhance scalability, system performance, and urban service optimization while ensuring secure and privacy-aware data handling.

Future work can focus on deploying the framework in real-world smart city environments using live IoT sensors and edge computing technologies. Advanced techniques such as federated learning, 5G integration, blockchain-based security, and intelligent decision-support systems can further improve scalability, privacy, and processing efficiency. The framework can also be extended to additional urban domains such as waste management, water distribution, and public safety systems.

The experimental evaluation confirms that the proposed framework efficiently processes high-volume urban data streams with minimal delay. The integration of machine learning models improves predictive analysis for traffic congestion, energy demand forecasting, and environmental anomaly detection. In addition, the distributed architecture ensures stable system performance even as the data volume increases, making the framework suitable for large-scale smart city applications.

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Appendices

Appendix A – System Architecture Diagram

The appendix includes the detailed architecture of the proposed real-time smart city analytics framework showing the integration of Apache Kafka, Spark Structured Streaming, Hadoop HDFS, MongoDB, machine learning models, dashboards, and alert systems.

Appendix B – Experimental Graphs and Results

This section contains graphical representations of:

- Accuracy comparison of machine learning models
- Processing latency analysis
- Throughput and scalability evaluation
- Domain-wise performance metrics
- Comparison with existing systems

Appendix C – Dataset Information

The appendix provides details of the datasets used in the study, including:

- Traffic dataset
- Energy consumption dataset
- Environmental monitoring dataset

It also includes preprocessing steps such as normalization, feature extraction, and missing value handling.

Appendix D – Tools and Configuration

This section includes the software tools, libraries, and parameter settings used for system implementation, including:

- Python 3.10

- Apache Kafka
- Apache Spark Structured Streaming
- Hadoop HDFS
- MongoDB
- Scikit-learn

Appendix E – Machine Learning Models

Detailed information about the machine learning models used in the framework:

- Random Forest for traffic prediction
- Linear Regression for energy forecasting
- Isolation Forest for anomaly detection

The appendix also includes model parameters and evaluation metrics.